Cover Letter — Electrical Engineering (Archiv für Elektrotechnik, Springer Nature)

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To: Prof. Dr. Igor Kuzle, Editor-in-Chief
Electrical Engineering (Archiv für Elektrotechnik), Springer Nature
Faculty of Electrical Engineering and Computing, University of Zagreb, Croatia

Subject: Submission of Original Research Article — "PSO-TCWavKAN: A Swarm-Optimized Temporal Convolutional Wavelet Kolmogorov-Arnold Network with Multi-Kernel Trend Decomposition and Asymptotic Linear Structural Recovery for Resilient Power Transformer Thermal Dynamics Forecasting"

Dear Prof. Dr. Kuzle and Esteemed Editorial Board,

I respectfully submit the above-titled original research manuscript for consideration in Electrical Engineering (Archiv für Elektrotechnik). This work presents PSO-TCWavKAN, a swarm-optimized temporal convolutional wavelet Kolmogorov-Arnold network designed for multi-step thermal dynamics forecasting in mineral-oil-immersed power transformers—a capability potentially relevant to substation dynamic thermal rating (DTR), condition-based asset management, and thermal overrun prevention (IEC 60076-7 / IEEE Std C57.91), subject to transformer-specific calibration and field validation.

Alignment with Scope & Relation to Prior Literature

This investigation directly addresses power equipment thermal modeling and builds upon the pioneering research of Bezerra et al. (2024), "A forecasting methodology based on second-generation wavelets and neural networks: application to power transformer oil temperature" (Electr Eng 106(3):3017–3029, DOI: 10.1007/s00202-023-02134-5). While Bezerra et al. established the utility of discrete wavelets with standard neural networks, our work substantially advances this methodology through three orthogonal contributions:

1. Continuous Dyadic Morlet Wavelet Filter Banks embedded in Kolmogorov-Arnold Network edges, providing continuous, differentiable multi-scale activations that eliminate discrete boundary distortion and avoid the training instability (Grid Update Dilemma) characteristic of discrete B-spline KANs;
2. Asymptotic Linear Structural Recovery (Proposition 1), providing a rigorous topological proof that the hypothesis-space closure contains the multi-kernel linear projection model (DLinear subspace) as routing gate weights approach zero, establishing representational protection against underfitting;
3. Hybrid CLPSO–PSO Warm-Start with Clerc constriction ($\chi = 0.7298$), initializing critical hyperparameters across a bounded 4-D search space on a fast surrogate partition to escape non-convex wavelet ripple basins prior to AdamW fine-tuning and Stochastic Weight Averaging (SWA).

Key Empirical Findings & Methodological Transparency

- Empirical Benchmarking: Evaluated across 16 canonical lookback–horizon configurations on standard ETT telemetry (ETTh1, ETTh2; $L \in \{96, 336\}$; $H \in \{96, 192, 336, 720\}$) under a continuous 70/10/20 chronological partition. DLinear, NLinear, and PSO-TCWavKAN are evaluated across 5 independent seeds (PatchTST Lite across 3 seeds). On volatile ETTh2, PSO-TCWavKAN outperforms all baselines in 7 of 8 configurations (achieving up to 13.6% MSE reduction over DLinear at H=720, and 8.5% for $H \leq 336$). Across all 16 configurations, it demonstrates nominal improvement over NLinear (uncorrected Wilcoxon $p = 0.0443$; Holm-corrected $p = 0.133$) and competitive parity with DLinear ($p = 0.274$; +1.4% higher mean MSE due to extended-horizon dispersion at H=720).
- Operational Robustness & Ablation: Controlled synthetic Gaussian noise injection ($\sigma = 0.30$) bounds MSE degradation between +1.4% and +13.0% across standard operational forecasting regime ($H \leq 336$). Systematic ablation across 240 runs confirms that every architectural module contributes meaningfully within the operational regime (omission increases error by +1.63% to +4.62%).
- Computational Economy: With 44.8 K parameters and 3.4 ms inference latency on NVIDIA Tesla T4, the architecture achieves a 3.8× parameter and 2.5× latency economy over patched Transformers.

Declarations & Open Science Commitment

In accordance with Springer Nature hybrid policies, this manuscript is submitted under the traditional subscription route (0 APC; The author has not selected Open Access). The paper is original, not submitted elsewhere, and has no competing interests or external funding. All code, benchmark CSV matrices, and reproduction scripts are openly accessible at: https://github.com/nphu211206/KAN-PSO-Transformer-Forecasting (MIT License). Extended logs and checkpoints are available upon request.

Respectfully yours,

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